CMPS 261 Progress Report: NASA Kepler Exoplanet Candidate Classification

Group 4: Joe Habre and Joy Habib Najem — **Track A:** 4.1 Tabular (NASA KOI)

GitHub Repository: `https://github.com/Joeehabre/CMPS-261`

1. Problem Statement

The NASA Kepler Objects of Interest (KOI) cumulative dataset contains $m = 9,564$ transit events observed by the Kepler Space Telescope across 153 archive attributes describing orbital parameters, transit signal geometry, photocenters, and host-star properties. Each observation $x^{(i)}$ is assigned an archive disposition (`koi_disposition`): `CONFIRMED` (2,748, 28.73%), `CANDIDATE` (1,977, 20.67%), or `FALSE POSITIVE` (4,839, 50.60%). Following our course learning framework—defining a hypothesis set $\mathcal{H}$ and learning algorithm $\mathcal{A}$ to approximate the target function $f : \mathcal{X} \to \mathcal{Y}$—our objective is to build a classifier that distinguishes promising exoplanet candidates from astrophysical and instrumental false positives (such as eclipsing binary stars or background stellar contamination). We formulate the primary task as binary classification: **Potential Exoplanet** (`CONFIRMED` ∪ `CANDIDATE`, $y^{(i)} = 1$, 4,725 instances, 49.40%) versus **False Positive** (`FALSE POSITIVE`, $y^{(i)} = 0$, 4,839 instances, 50.60%). Because unconfirmed candidates are provisional archive dispositions that may contain residual label noise, we also define a secondary strict benchmark restricted to verified dispositions (`CONFIRMED` vs. `FALSE POSITIVE`, $m = 7,587$) to evaluate label-noise sensitivity.

Changes Since Initial Proposal: While our core problem formulation remains unchanged from the initial proposal, we introduced two methodological refinements during baseline development: (1) an explicit feature audit that removes direct post-classification disposition scores and false-positive flags (`koi_fpflag_*`), and (2) **host-star grouped partitioning** (`GroupShuffleSplit` and `GroupKFold` on star ID `kepid`) so multi-candidate stellar systems never straddle training and validation splits.

2. Data Preprocessing

1. **Target Leakage Audit & Feature Filtering:** Several attributes in the cumulative KOI archive are produced *after* human or pipeline dispositioning and directly leak the target label. Using archive documentation, we audited all columns and removed:
   - *Direct Target & Disposition Columns (7):* `koi_disposition`, `koi_pdisposition` (preliminary pipeline disposition), `koi_score` (vetting confidence score in $[0, 1]$), and the four false-positive flags: `koi_fpflag_nt` (not transit-like), `koi_fpflag_ss` (stellar eclipse), `koi_fpflag_co` (centroid offset), and `koi_fpflag_ec` (ephemeris match contamination).
   - *Identifiers & Categorical Provenance Metadata (17):* `rowid`, `kepid` (retained solely as a grouping key), `kepoi_name`, `kepler_name` (assigned only to confirmed planets, directly leaking $y = 1$), `koi_tce_delivname`, and 12 archive provenance/comment strings.
   - *High-Missingness Columns ($\geq$ 50% NaN on Development Set):* Computed missingness rates strictly on the development partition ($m_{\text{dev}} = 7,662$) and dropped sparsely populated columns (such as `koi_teq_err1`, `koi_teq_err2`, and `koi_sage`), retaining 103 numerical attributes. Note that while direct disposition flags are removed, several retained columns are pipeline centroid/vetting diagnostic statistics (`koi_dicco_*`, `koi_fwm_*`); we will quantify their prediction-time availability via feature ablation in Milestone 2.
2. **Domain Feature Engineering & Skewness Transformation:** To help linear and neural models capture nonlinear physical laws without high-degree polynomial explosion, we engineered 8 domain-specific features ($n = 111$ total features): (i) *Geometric Transit Consistency* comparing observed transit depth δ (`koi_depth`) against the expected planet-to-star area ratio $(R_p/R_*)^2$, as grazing eclipsing binaries violate $\delta \approx (R_p/R_*)^2$; (ii) *Transit Duty Cycle* (`koi_duration`/`koi_period`); and (iii) $\log(1+x)$ transformations on six heavily right-skewed positive variables (`koi_period`, `koi_depth`, `koi_prad`, `koi_insol`, `koi_model_snr`, `koi_duration`).
3. **Grouped Splitting, Imputation, and Standardization:** Because 9,564 KOIs orbit 8,214 unique host stars (`kepid`), random row splitting leaks stellar properties (`koi_steff`, `koi_srad`) between train and test sets. We partitioned the dataset into an 80% development set ($m_{\text{dev}} = 7,662$ KOIs) and a 20% held-out test set ($m_{\text{test}} = 1,902$ KOIs) grouped by `kepid` (0 shared stars), leaving the test set completely untouched. Within 5-fold `GroupKFold` cross-validation on the development set, missing entries (363 rows missing `koi_srad` overall; 288 in m_{dev}) are imputed inside each training fold via **Median Imputation** plus binary missingness indicators (`add_indicator=True`), followed by **Z-score Standardization** ($z = \frac{x-\mu}{\sigma}$) for linear and neural models. For XGBoost, we pass raw features directly to leverage native missing-value routing.

3. Evaluation Metrics

- **ROC-AUC and PR-AUC / Average Precision (Ranking Performance):** Because follow-up radial-velocity spectroscopy is resource-constrained, astronomers prioritize top-ranked candidates by predicted probability. ROC-AUC evaluates global ranking discrimination via True Positive Rate ($\text{TPR} = \frac{TP}{TP+FN}$) vs. False Positive Rate ($\text{FPR} = \frac{FP}{FP+TN}$), while Precision-Recall evaluation (computed as Average Precision, denoted PR-AUC) quantifies positive-class precision across recall thresholds.
- **F1-Score, Precision, and Recall (Thresholded Classification):** At decision threshold $\tau = 0.5$, False Negatives discard viable exoplanet candidates whereas False Positives waste telescope time on stellar binaries. Reporting $\text{Precision} = \frac{TP}{TP+FP}$, $\text{Recall} = \frac{TP}{TP+FN}$, and their harmonic mean ($\text{F1} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision}+\text{Recall}}$) quantifies both error types.
- **Expected Calibration Error (Confidence Trustworthiness):** Because raw model scores in $(0, 1)$ are not automatically calibrated, we compute 10-bin Expected Calibration Error ($\text{ECE} = \sum_{b=1}^{10} \frac{|B_b|}{m} |\text{acc}(B_b) - \text{conf}(B_b)|$) and plot reliability diagrams to assess whether predicted confidence matches empirical accuracy.

4. Classical Baseline (Level 1)

We implemented a structured Level 1 classical baseline ladder evaluated via 5-fold `GroupKFold` cross-validation on the development set:

1. **Tuned Linear Baseline (Regularized Logistic Regression):** Linear logit $z = w \cdot x + b$ with Sigmoid activation $f_{w,b}(x) = g(z) = \frac{1}{1+e^{-z}}$, trained under Binary Cross-Entropy loss with L_2 regularization. Using a 3-fold `GroupKFold` validation search over $C \in \{0.01, 0.1, 1.0, 10.0\}$, mean ROC-AUC progressed from 0.9469 ($C = 0.01$) and 0.9489 ($C = 0.1$) to a peak of 0.9496 at $C^* = 1.0$ (vs. 0.9494 at $C = 10.0$), selecting $C^* = 1.0$.
2. **Bagging Baseline (Random Forest):** A fixed-configuration ensemble of $B = 500$ decision trees trained via bootstrap sampling and random feature subsets $K = \sqrt{n}$ at each split (`max_depth=16`) to decorrelate trees and reduce variance.
3. **Boosted Baseline (XGBoost):** Sequential gradient-boosted decision trees trained on first-order gradients (g_i) and second-order Hessians (h_i) of the log-loss objective, optimizing regularized split gain $\text{Gain} = \frac{1}{2}\left[\frac{G_L^2}{H_L+\lambda} + \frac{G_R^2}{H_R+\lambda} - \frac{(G_L+G_R)^2}{H_L+H_R+\lambda}\right] - \gamma$ under a fixed baseline configuration (500 trees, learning rate $\eta = 0.03$, `max_depth`= 6, L_2 regularization $\lambda = 5.0$, split penalty $\gamma = 0.5$, row/column subsampling = 0.8).

Model (5-Fold Host-Star Grouped CV Mean ± SD)	ROC-AUC	PR-AUC (AP)	F1-Score	Precision	Recall	ECE (↓)	Fit Time/Fold
Logistic Regression (Tuned $L_2, C = 1.0$)	0.9523 ± 0.0073	0.9485 ± 0.0089	0.8845 ± 0.0166	0.8708	**0.8989**	0.0327	1.25 s
Random Forest (500 trees, `max_depth=16`)	0.9684 ± 0.0045	0.9704 ± 0.0051	$\mathbf{0.9006 \pm 0.0106}$	**0.9181**	0.8839	0.0465	18.10 s
XGBoost (Classical Baseline)	$\mathbf{0.9711 \pm 0.0032}$	$\mathbf{0.9723 \pm 0.0042}$	0.8987 ± 0.0120	0.9058	0.8919	0.0262	13.62 s
ResNet-MLP (Temp. Scaled, Level 2 Prelim.)	0.9659 ± 0.0046	0.9676 ± 0.0054	0.8909 ± 0.0090	0.9065	0.8760	**0.0211***	14.54 s

5. Sophisticated Model (Level 2)

- **Architecture & Justification:** For Level 2, we implemented a deep tabular **Residual Multi-Layer Perceptron (ResNet-MLP)** with residual skip connections ($y = F(x) + x$), Batch Normalization, ReLU activations, Dropout ($p = 0.25$), and Sigmoid output, trained with mini-batch AdamW ($\alpha = 10^{-3}$, weight decay 10^{-4}, batch size 128) for 50 epochs per fold while retaining the checkpoint with the lowest validation loss. While decision trees partition continuous orbital laws ($P^2 \propto a^3/M_*$) via axis-aligned step functions, a residual network learns smooth hierarchical representations across coupled orbital and stellar features. We also explored post-hoc **Temperature Scaling** ($\hat{p} = \sigma(z/T)$) on frozen logits.
- **Current Progress & Preliminary Results:** ResNet-MLP substantially outperforms tuned Logistic Regression (+0.0136 mean ROC-AUC, +0.0191 mean PR-AUC), confirming effective non-linear representation learning. In raw ranking metrics, standalone ResNet-MLP slightly trails XGBoost (0.9659 vs. 0.9711 mean ROC-AUC) with comparable CPU fit time (14.54 s vs. 13.62 s per fold, $\sim 1.07\times$). *Note on Preliminary Calibration:* In our current notebook, best-epoch selection uses the validation fold and T is fitted on the first half of each validation fold before scoring the full fold (yielding 0.0211 fold-average ECE and 0.008 pooled OOF ECE). Because calibration and evaluation examples overlap within the validation fold, we treat this calibration result as preliminary and will implement a strictly disjoint three-way split (train / inner-validation & calibration / outer-evaluation) in Milestone 2.

6. Remaining Plan and Risks (Until November 20)

- **Remaining Experiment Schedule:**
  - *Nov 6–Nov 10 (Disjoint Calibration, Level 2 Tuning & Ensembling):* Implement strict nested validation so early stopping and Temperature Scaling (T) use held-out inner splits completely separate from evaluation folds; tune XGBoost, ResNet-MLP, and FT-Transformer architectures and select tree + neural ensembles strictly on development cross-validation data.
  - *Nov 11–Nov 14 (Split Sensitivity & Feature Provenance Ablation):* Using development CV, compare (i) Host-Star Grouped Split vs. Random Split (quantifying host-star leakage), (ii) Candidate grouping vs. Strict Confirmed-only disposition, and (iii) an ablation isolating core physical transit/stellar parameters from pipeline vetting diagnostics (`koi_dicco_*`, `koi_fwm_*`).
  - *Nov 15–Nov 20 (Final Test Evaluation, Complexity Trade-offs & Deliverables):* Evaluate locked final models and calibration once on the untouched held-out test set ($m_{\text{test}} = 1,902$); benchmark latency/complexity trade-offs and finalize the report and poster.
- **Early Observations on Model Failures:** Automated subgroup analysis of out-of-fold XGBoost predictions across the 7,662 development instances confirms systematic failure regimes: (1) **Weak Signals (`koi_model_snr` < 12, 1,816 dev cases):** error rate reaches 21.15%, nearly 5× higher than high-SNR transits (SNR > 50, 2,310 dev cases, 4.29% error); (2) **Missing Stellar Radius (288 dev cases missing `koi_srad` out of 363 total):** error rate jumps to 28.47% due to unresolvable planet-to-star radius ratios; and (3) **Grazing Eclipsing Binaries (`koi_impact` > 0.9) & Evolved Stars (`koi_slogg` < 3.8):** V-shaped eclipses and diluted giant-star transits mimic planetary signals.
- **Risks & Mitigations:** Deep tabular networks risk overfitting noisy features and trailing boosted trees in raw AUC. Enforcing strict held-out calibration splits, evaluating tree + neural ensembles on development folds, and quantifying feature-provenance ablations directly addresses these methodological risks.

Appendix: Preliminary Experimental Figures (Excluded from 2-Page Text Limit)

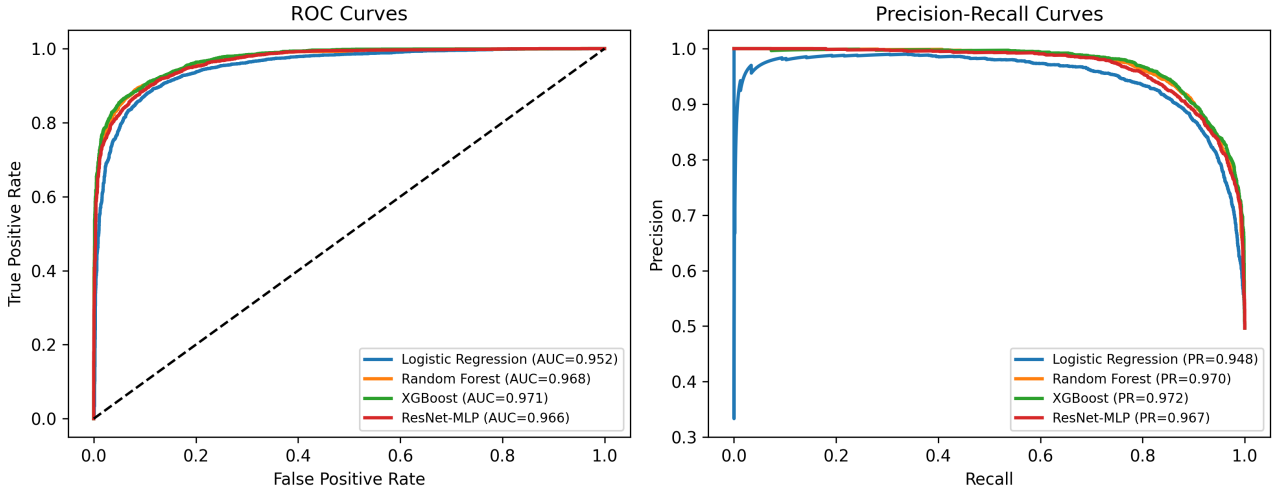


Figure 1: 5-Fold Host-Star Grouped Cross-Validation pooled out-of-fold ROC (left) and Precision-Recall (right) curves comparing Level 1 classical baselines (Tuned Logistic Regression: blue, Random Forest: orange, XGBoost Baseline: green) against the Level 2 ResNet-MLP (red). Pooled OOF AUC/PR values in the legend closely match the fold-average metrics in Table 1.

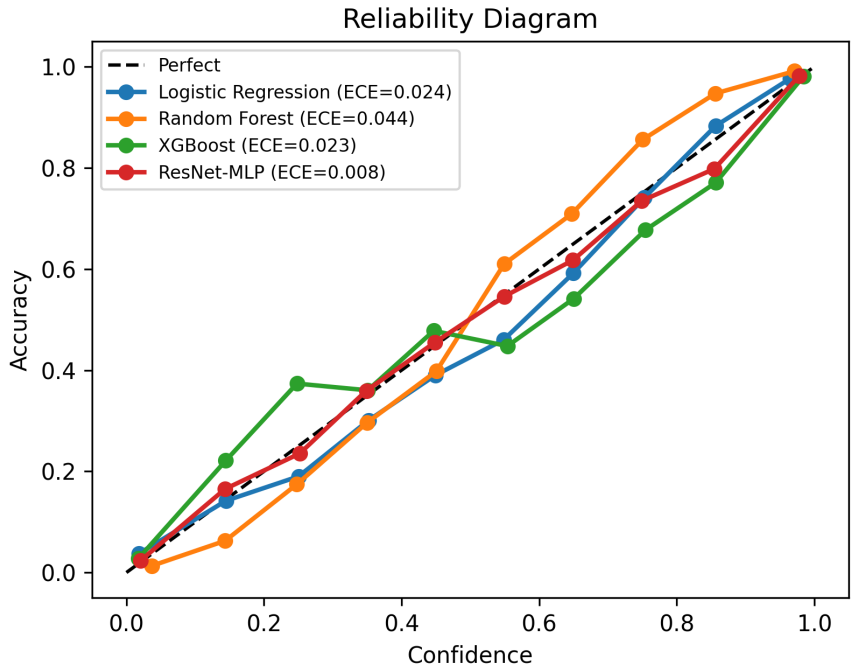


Figure 2: Reliability diagram across 10 probability bins on 5-fold grouped cross-validation. Legend values report pooled out-of-fold ECE ($ECE_{pooled} = 0.024$ for Logistic Regression, 0.044 for Random Forest, 0.023 for XGBoost, and 0.008 for ResNet-MLP), whereas Table 1 reports fold-average ECE (0.0327, 0.0465, 0.0262, and 0.0211, respectively). Because ResNet-MLP epoch selection and temperature scaling (T) were fitted within validation folds, ResNet-MLP calibration is preliminary pending disjoint held-out calibration in Milestone 2.